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Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 49901Roll
No.**B. Tech. Examination, 2025-26****(Special Carry Over Paper)****MATRICES AND CALCULUS****Time : Three Hours]****[Maximum Marks : 60****Note :—** Attempt all questions.**SECTION – A**1. Attempt all parts of the following : $8 \times 1 = 8$

- (a) Define identity matrix of order 2×2 .
- (b) Write the statement of Cayley-Hamilton theorem.
- (c) Evaluate $D^n (e^{ax})$.
- (d) Define :

$$\frac{\partial (u, v)}{\partial (x, y)}$$

[P. T. O.]

- (e) What do you mean by double integral?
- (f) Write the value of $\frac{1}{2}$.
- (g) If $\vec{r} = t^2 \hat{i} + 2t \hat{j} + \hat{k}$, then find the value of $\frac{d\vec{r}}{dt}$.
- (h) Write the statement of Gauss divergence theorem.

SECTION-B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Find the rank of the matrix :

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$$

- (b) If $y = \sin(m \sin^{-1} x)$, prove that :

$$(1 - x^2) y_{n+2} - (2n + 1) x y_{n+1} + (m^2 - n^2) y_n = 0$$

- (c) Evaluate :

$$\int_0^1 \int_0^y x y e^{-x^2} dx dy$$

- (d) Use Green's theorem to evaluate :

$$\int_C (x^2 + xy) dx + (x^2 + y^2) dy$$

where C is the square formed by the lines $y = \pm 1, x = \pm 1$.

SECTION-C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) Determine for what values of λ and μ the following equations :

$$\begin{aligned} x + y + z &= 6, \\ x + 2y + 3z &= 10, \\ x + 2y + \lambda z &= \mu \end{aligned}$$

have :

- (i) No solution
(ii) A unique solution
(iii) Infinite number of solutions

[P. T. O.]

- (b) Find the eigen values and corresponding eigen vectors for the matrix :

$$A = \begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$$

- (c) Verify Cayley-Hamilton theorem for the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

and hence find A^{-1} .

4. (a) If:

$$u = \log \left(\frac{x^2 + y^2}{x + y} \right)$$

prove that :

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 1$$

- (b) Expand $\sin x$ in power series.
- (c) Use the method of the Lagrange's multipliers to find the volume of the largest rectangular parallelopiped that can be inscribed in the ellipsoid :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

5. (a) Evaluate the integral :

$$\int_0^\infty \int_0^x x \exp \left(\frac{-x^2}{y} \right) dx dy$$

by changing the order of integration.

- (b) Evaluate :

$$\int_0^{\pi/2} \sin^2 x \cos^3 x dx$$

- (c) Find the mass of an octant of the ellipsoid :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

the density at any point being $\rho = Kxyz$.

6. (a) Find the curl of:

$$\vec{v} = (xyz) \hat{i} + (3x^2y) \hat{j} + (xz^2 - y^2z) \hat{k}$$

at the point $(2, -1, 1)$.

- (b) If $\vec{F}(x, y, z) = x^3 \hat{i} + y \hat{j} + z \hat{k}$ is the forcefield.

Find the workdone by $\vec{F}$ along the line from the $(1, 2, 3)$ to $(3, 5, 7)$.

[P. T. O.]

(c) Evaluate $\int \int_s \vec{F} \cdot \hat{n} \, ds$ where:

$$\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$$

and s is the surface of the cube bounded by:

$$x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$$
